

Fluid Flow Operation

Text Books:

- **K.A. Gavanhe, “Unit Operations - I”, Nirali Prakashan, 2015**
- W. L. McCabe, J.C. Smith and P. Harriott, “Unit operations of Chemical Engineering”, McGraw Hill, International Edn., 7th Edition, 2005.

Unit 4: Flow of Fluids through Solids

- Form drag, skin drag, Drag coefficient
- Flow around solids and packed beds
- Friction factor for packed beds
- Ergun's Equation
- Motion of particles through fluids
- Terminal settling velocity.
- Fluidisation - Mechanism, types, general properties, applications

Pressure Drop In Flow Through Porous Media

- Porous medium : It is a continuous solid phase with many pores or void spaces in it.
- Typical examples of porous media include sponges, filters and packed beds used for absorption and distillation, etc.
- Porous media which are not having interconnected pores are said to be impermeable to fluid flow.
- In such cases, there is no possibility of a fluid to flow through them, e.g., foamed polystyrene drinking cups.
- Porous media which are having interconnected pores are said to be permeable to fluid flow. In such cases, there is a flow of fluid through them, e.g., a bed of granular solids.
- The flow of fluids through permeable porous media is of great practical importance in filters, packed columns used for absorption and distillation and fixed-bed catalytic reactors.

Pressure Drop In Flow Through Porous Media

- Consider the flow of a homogeneous fluid through a bed of spherical particles of uniform size.
- If the average velocity at any cross-section perpendicular to the flow is based on the entire cross-sectional area of the pipe/tower/bed (i.e., empty tower/bed cross-section), then it is called as the superficial velocity and is given by

$$u_o = Q/A$$

- The average velocity may be based on the area actually available for the flowing fluid, in which case it is called as the interstitial velocity.
- where e is the porosity, or void fraction or voidage of the bed.

$$u = \frac{u_o}{e} = \frac{Q}{e.A}$$

Pressure Drop In Flow Through Porous Media

e is the porosity, or void fraction or voidage of the bed.

$$e = \frac{\text{total volume of bed} - \text{volume of solids in bed}}{\text{total volume of bed}}$$
$$= \frac{\text{free volume available for flow}}{\text{total volume of fixed bed}}$$

- If D is the diameter of a pipe/bed/tower, h is the height of a bed in the pipe/tower, n is the number of solids in the bed and v_p is the volume of one solid particle then

$$e = \frac{\pi/4 D^2 h - n v_p}{\pi/4 D^2 h}$$

Pressure Drop In Flow Through Porous Media

- For a packed bed, the hydraulic radius varies along the bed height and is given by

$$r_H = \frac{\text{Wetted cross-sectional area normal/perpendicular to fluid flow}}{\text{Wetted perimeter}}$$

- Multiplying both the numerator and the denominator by the length of the bed / the height of the bed

$$r_H = \frac{\text{Volume open to flow/available for flow}}{\text{Total wetted surface area of packing}}$$

- For a porous medium made of spherical particles of uniform size, the hydraulic radius is given by

$$r_H = \frac{\text{Total volume of bed} \times e}{\text{Number of particles} \times \text{Surface area of one particle}}$$

Pressure Drop In Flow Through Porous Media

- The number of solid particles in the bed are given by

$$\text{Number of particles} = \frac{\text{Volume of bed} \times (1 - e)}{\text{Volume of one particle}}$$

- Thus hydraulic radius becomes, where d_p is the diameter of the spherical particle

$$\begin{aligned} r_H &= \frac{e}{(1 - e) \times \left(\frac{\text{surface area}}{\text{volume}} \right) \text{ of a spherical particle}} \\ &= \frac{e}{(1 - e) (\pi d_p^2 / \pi/6 d_p^2)} \end{aligned}$$

$$r_H = \frac{d_p}{6} \left(\frac{e}{1 - e} \right)$$

Pressure Drop In Flow Through Porous Media

- For flow through packed beds, the Reynolds number and the friction factor become

$$\begin{aligned} N_{Re} &= D_e u \rho / \mu \\ &= (4 r_H) (u_o / e) \rho / \mu \end{aligned}$$

- Substituting for r_H gives

$$\begin{aligned} N_{Re} &= 4 \frac{d_p}{6} \times \frac{e}{1-e} \times \frac{u_o}{e} \times \frac{\rho}{\mu} \\ N_{Re} &= \frac{2 d_p u_o \rho}{3 \mu (1-e)} \end{aligned}$$

Pressure Drop In Flow Through Porous Media

- We know,

$$\frac{\Delta P}{\rho} = \frac{4 f L u^2}{2 D}$$

- Replacing D by D_e and rearranging, we get

$$f = \left(\frac{\Delta P}{\rho L} \right) \left(\frac{D_e}{2 u^2} \right)$$

- Replacing D_e by $4 r_H$ gives

$$f = \left(\frac{\Delta P}{\rho L} \right) \frac{(4 r_H)}{2 (u_o/e)^2}$$

- Substituting for r_H gives

$$f = \left(\frac{\Delta P}{\rho L} \right) 4 \frac{d_p}{6} \times \frac{e}{1-e} \times \frac{e^2}{2 u_0^2}$$

$$f = \frac{1}{3} \left(\frac{\Delta P}{\rho L} \right) \frac{d_p}{u_0^2} \times \frac{e^3}{1-e}$$

Pressure Drop In Flow Through Porous Media

- Hence, the modified friction factor and the Reynolds number for flow through packed beds are defined as

$$f_m = \left(\frac{\Delta P}{\rho L} \right) \frac{d_p}{u_o^2} \times \frac{e^3}{1 - e}$$

$$N_{Re,,m} = \frac{d_p u_o \rho}{\mu (1 - e)}$$

Pressure Drop In Flow Through Porous Media

- The Ergun equation for the flow of homogeneous fluids through packed beds relating f_m and $N_{Re,m}$ is

$$f_m = \frac{150}{N_{Re,m}} + 1.75$$

- Equation fits experimental data well for $N_{Re,m}$ between 1 and 2000.

Pressure Drop In Flow Through Porous Media

- For $N_{Re,m} < 10$ (i.e., when the flow is laminar/streamline through packed beds), the viscous forces control and Equation reduces to

$$f_m = \frac{150}{N_{Re,m}}$$

- Substituting the values of f_m and $N_{Re,m}$

$$\left(\frac{\Delta P}{\rho L}\right) \frac{d_p}{u_0^2} \times \frac{e^3}{1-e} = \frac{150}{d_p u_0 \rho / \mu (1-e)}$$

$$\left(\frac{\Delta P}{\rho}\right) = \frac{150 \mu u_0 L}{d_p^2 \rho} \times \frac{(1-e)^2}{e^3}$$

- Equation is known as the Blake-Kozeny equation, or Kozeny-Carman equation.
- The flow of fluids through industrial filters is usually streamline/laminar and thus Equation applies.

Pressure Drop In Flow Through Porous Media

- For $N_{Re,m} > 1000$ (i.e., when the flow is turbulent), the effect of viscous forces is negligible and inertial forces control and Equation reduces to

$$f_m = 1.75$$

$$\frac{\Delta P}{\rho} = 1.75 \frac{u_0^2 L}{d_p} \times \frac{(1 - e)}{e^3}$$

- Equation is known as the Burke-Plummer equation.

Pressure Drop In Flow Through Porous Media

- **For non-spherical particles**, the diameter d_v of a non-spherical particle may be defined as the diameter of a sphere having the same volume as that of the particle.
- For non-spherical particles, the shape factor or sphericity ϕ_s is defined as below
- For spheres : $\phi_s = 1$ and for all other particle shapes : $0 < \phi_s < 1$
- In equation, $dp = \Phi_s d_v$
- To obtain **Ergun Equation**

$$r_H = \frac{d_p}{6} \left(\frac{e}{1-e} \right) \quad \frac{\Delta P}{\rho} = 1.75 \frac{u_0^2 L}{d_p} \times \frac{(1-e)}{e^3}$$

$$\frac{\Delta P}{\rho L} = 150 \frac{\mu u_0}{\rho (\phi_s d_v)^2} \frac{(1-e)^2}{e^3} + 1.75 \frac{u_0^2}{\phi_s d_v} \times \frac{1-e}{e^3}$$

Fluidisation

- When a fluid (gas or liquid) is passed up through a bed of solid particles at very low velocity, the particles do not move and the pressure drop is given by the Ergun equation.
- If the fluid velocity is steadily increased, the pressure drop and drag on the individual particles increase, and eventually the particles start to move and become suspended in the fluid stream.
- An operation by which fine granular solids are transformed into a fluid-like state through contact with a fluid is known as fluidisation.
- The term fluidised bed is used to describe the condition of fully suspended particles in a fluid stream.

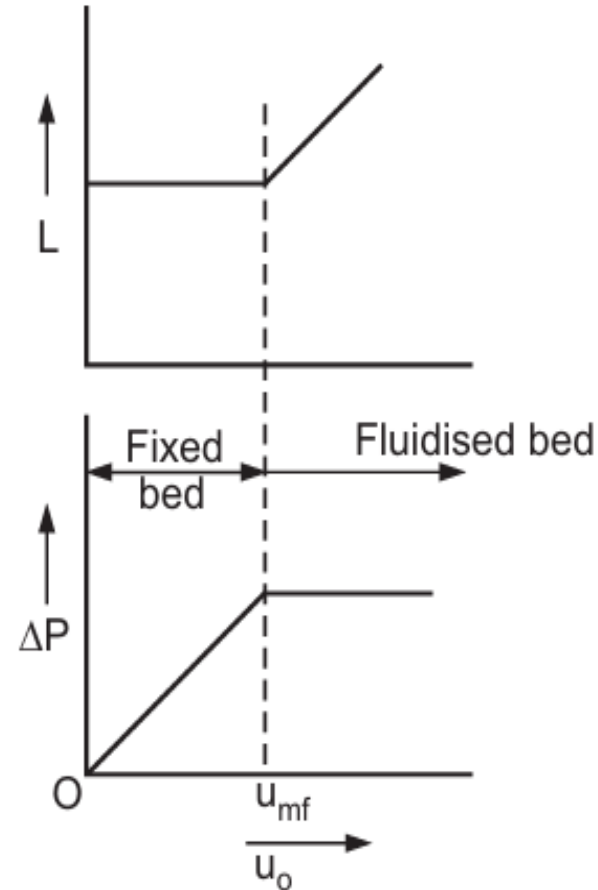
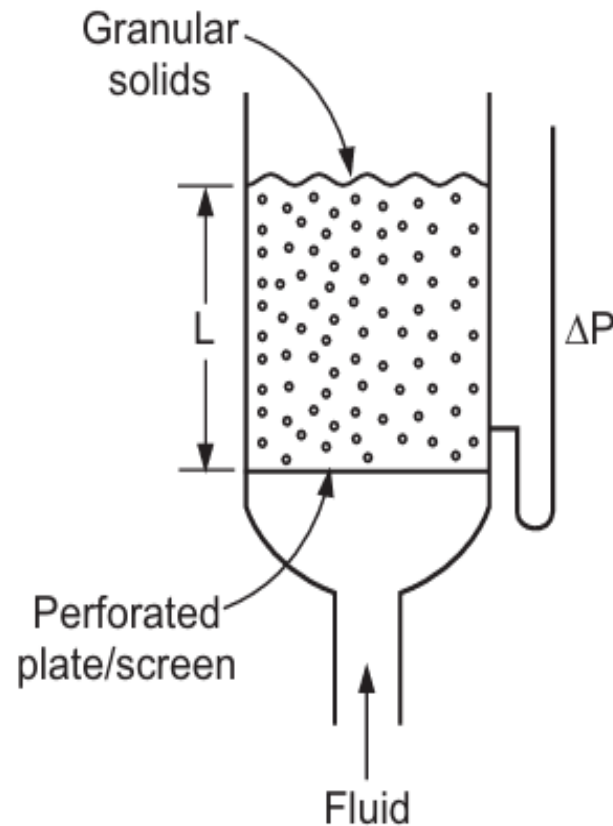
Advantages of fluidisation

- (i) good contact of the fluid with all parts of the solid particles
 - (ii) prevents segregation of the solid particles by thoroughly agitating the bed
 - (iii) temperature is uniform even in a large reactor due to complete mixing
 - (iv) high heat and mass transfer rates.
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- Fluidisation finds applications in catalytic cracking, drying and transportation of solids, etc.

Disadvantages of fluidisation

- (i) greater power requirement
- (ii) higher breakage of solid particles
- (iii) serious erosion of pipelines and vessels
- (iv) necessity of recovery systems.

Fluidisation



- Consider a vertical tube partially filled with a fine granular particles.
- The tube is open at the top and has a porous plate to support the bed of particles and to distribute the flow uniformly over the entire cross-section.

Fluidisation

- Further, consider the upward flow of a fluid through a vertical bed of fine particles.
- At a low flow rate, the fluid simply passes through the void spaces between stationary particles without causing any particle motion.
- The flow will be laminar and the pressure drop across the bed will be proportional to the superficial velocity u_0 .
- As we go on increasing steadily the flow rate of the fluid, a stage will be reached when all the particles are just suspended in the flowing fluid.
- At this stage, the pressure drop across the bed equalizes the weight of the bed.
- The bed in this condition is considered to be just fluidised and is termed as an incipiently fluidised bed.